GenAI-Based Cybersecurity Incident Investigation Report

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1. Incident Summary

Investigation Metric	Value
Total Log Events	8
Risk Score	10/10
Severity	HIGH
Possible Root Cause	Possible Compromised User Credentials
Root Cause Confidence	High

2. Severity Assessment

• Repeated failed login attempts detected.

• Successful authentication occurred after suspicious login activity.

• Access from an unknown device was detected.

• Sensitive resources were accessed.

• Sensitive file download activity was detected.

3. Root Cause Analysis

Possible Root Cause: Possible Compromised User Credentials

Confidence: High

Reasoning: The available evidence shows a sequence of security events that may be connected. The observed authentication activity, device information, and resource access patterns support the proposed root-cause hypothesis.

Supporting Evidence:

• 3 failed login attempts were detected before successful authentication.

• Successful authentication was observed from an IP address different from the previous failed-login activity.

• Access from an unknown device was detected.

• 3 sensitive resource access event(s) were detected.

• A file download event was detected after suspicious authentication activity.

4. Recommended Defensive Actions

Priority	Category	Recommended Action
HIGH	Account Security	Review the affected account's authentication activity.
HIGH	Authentication Review	Verify whether the successful login was performed by the legitimate user.
HIGH	Device Investigation	Verify the identity and ownership of the unknown device.

Priority	Category	Recommended Action
HIGH	Network Investigation	Review the source IP addresses associated with the suspicious authentication events.
HIGH	Data Protection	Review access permissions and access history for the affected sensitive resources.
MEDIUM	Evidence Preservation	Preserve relevant file, database, and access logs for further investigation.
HIGH	Data Exposure Investigation	Review the downloaded files and determine whether the transfer was authorized.
HIGH	Credential Protection	Consider credential reset and verify multi-factor authentication status according to organizational policy.
MEDIUM	Endpoint Investigation	Review endpoint security logs and device activity associated with the incident.
MEDIUM	Forensic Investigation	Preserve the original logs and maintain an investigation record before making changes to affected systems.
MEDIUM	Continuous Monitoring	Monitor the affected account, devices, IP addresses, and sensitive resources for additional suspicious activity.
LOW	Investigation Closure	Verify all findings with additional authentication, endpoint, network, and application evidence before closing the in

5. GenAI Investigation Report

1. Incident Overview

The supplied logs show a suspicious authentication and access sequence involving the `admin` account. Three failed `admin` logins occur from `192.168.1.20`, followed by a successful `admin` login from a different source IP, `185.220.101.45`. That successful session is then associated with access to an `Unknown Device`, the file `Employee_Salaries.xlsx`, the `Employee Database`, and a download of `Employee_Salaries.xlsx`.

Confirmed observation: the log events above occurred in the order and with the statuses shown.

Investigative hypothesis: this may represent unauthorized account access or account compromise. The evidence is suspicious and consistent with that possibility, but it does not by itself prove how access was obtained, whether the activity was authorized, or whether data exfiltration occurred.

2. Observed Suspicious Activity

- Three failed `admin` login attempts from `192.168.1.20`:

- `2026-10-03 09:00:12`

- `2026-10-03 09:01:05`

- `2026-10-03 09:02:18`

- Evidence: all are `Login` events for user `admin`, resource `User Account`, status `Failed`.

- Successful `admin` login from a different source IP, `185.220.101.45`, at `2026-10-03 09:04:31`.

- Evidence: `Login` event for `admin`, resource `User Account`, status `Successful`.

- Suspicious because the source IP differs from the IP used in the preceding failed attempts.

- Immediate post-login activity from `185.220.101.45`:

- `09:05:10` — `Device Access` to `Unknown Device`, status `Detected`.

- `09:06:45` — `File Access` to `Employee_Salaries.xlsx`, status `Accessed`.

- `09:08:22` — `Database Access` to `Employee Database`, status `Accessed`.

- `09:10:14` — `File Download` of `Employee_Salaries.xlsx`, status `Downloaded`.

- Evidence: all events are associated with user `admin` and source IP `185.220.101.45`.

- The accessed resources appear sensitive based on their names: `Employee_Salaries.xlsx` and `Employee Database`.

- Evidence: the log resource names explicitly include employee salary and employee database references.

3. Evidence Correlation

- Same user across all events: every event lists user `admin`.

- Evidence: all eight log entries have `"user": "admin"`.

- Source IP change after failed logins:

- Failed attempts: `192.168.1.20`.

- Successful login and subsequent activity: `185.220.101.45`.

- Evidence: the first three events use `192.168.1.20`; the successful login and all later events use `185.220.101.45`.

- Temporal sequence is tight and ordered:

- Failed logins: `09:00:12` to `09:02:18`.

- Successful login: `09:04:31`.

- Device access: `09:05:10`.

- File access: `09:06:45`.

- Database access: `09:08:22`.

- File download: `09:10:14`.

- Evidence: timestamps show a progression from authentication failure to successful authentication to sensitive resource access and download.

- Confirmed observation: the successful login and subsequent sensitive access came from `185.220.101.45`, not from the IP used in the failed attempts.

- Investigative hypothesis: the failed attempts from `192.168.1.20` may be related to the later successful login from `185.220.101.45`, but the logs do not prove a causal link. They could be separate events, prior failed attempts, or activity from a different actor or system.

4. Attack / Incident Timeline

| Timestamp | User | Source IP | Event Type | Resource | Status | Evidence-Based Interpretation |

|---|---|---|---|---|---|---|

| 2026-10-03 09:00:12 | admin | 192.168.1.20 | Login | User Account | Failed | Failed admin login from `192.168.1.20`. |

| 2026-10-03 09:01:05 | admin | 192.168.1.20 | Login | User Account | Failed | Second failed admin login from same IP. |

| 2026-10-03 09:02:18 | admin | 192.168.1.20 | Login | User Account | Failed | Third failed admin login from same IP. |

| 2026-10-03 09:04:31 | admin | 185.220.101.45 | Login | User Account | Successful | Successful admin login from a different source IP. |

| 2026-10-03 09:05:10 | admin | 185.220.101.45 | Device Access | Unknown Device | Detected | Admin session accesses an unknown device. |

| 2026-10-03 09:06:45 | admin | 185.220.101.45 | File Access | Employee_Salaries.xlsx | Accessed | Admin session accesses a salary-related file. |

| 2026-10-03 09:08:22 | admin | 185.220.101.45 | Database Access | Employee Database | Accessed | Admin session accesses an employee database. |

| 2026-10-03 09:10:14 | admin | 185.220.101.45 | File Download | Employee_Salaries.xlsx | Downloaded | Admin session downloads the salary-related file. |

5. Possible Root Cause

Confirmed observation: the root cause is not established by the supplied evidence.

Investigative hypotheses:

- Compromised `admin` credentials:

- Supported by: successful `admin` login from `185.220.101.45` immediately after three failed `admin` logins from `192.168.1.20`, followed by sensitive resource access and download.

- Not proven by: the logs do not show how credentials were obtained, whether MFA was used, or whether the login was authorized.

- Password guessing or brute-force activity:

- Supported by: three failed `admin` logins from `192.168.1.20`.

- Not proven by: only three failed attempts are shown, which is limited evidence for brute force. The logs do not show repeated high-volume attempts or lockout events.

- Unauthorized remote access from `185.220.101.45`:

- Supported by: successful login and all subsequent sensitive activity originate from that IP.

- Not proven by: the logs do not identify whether `185.220.101.45` is a known corporate VPN, proxy, admin location, or malicious source.

- Legitimate administrator activity:

- Possible because the login is marked `Successful` and the user is `admin`.

- However, the sequence of failed logins, source IP change, unknown device access, and sensitive file/database access makes this suspicious. The logs do not show explicit authorization or business justification.

- Session hijacking, token theft, or insider misuse:

- Not directly supported by the supplied evidence. These remain hypotheses only if additional logs confirm them.

6. Risk and Impact

Confirmed potential impact based on evidence:

- The `admin` account successfully authenticated from `185.220.101.45`.

- The same session accessed an `Unknown Device`.

- The same session accessed `Employee_Salaries.xlsx`.

- The same session accessed the `Employee Database`.

- The same session downloaded `Employee_Salaries.xlsx`.

Risk implications:

- High potential confidentiality risk because employee salary data and an employee database were accessed.

- Potential data exfiltration risk because `Employee_Salaries.xlsx` was downloaded. The logs confirm a download event, but they do not confirm where the file was transferred or whether it left the organization.

- Administrative account compromise could allow broader access, persistence, or lateral movement, but the supplied logs do not show those events.

- Regulatory, privacy, and compliance risk may exist if employee data was accessed or disclosed without authorization. The logs do not establish legal or compliance conclusions.

Not confirmed by evidence:

- Actual data exfiltration outside the environment.

- Malware execution.

- Lateral movement.

- Persistence mechanisms.

- Whether `185.220.101.45` is malicious.

- Whether the `admin` user authorized the activity.

7. Recommended Investigation Steps

- Preserve all relevant logs and evidence, including authentication, device, file, database, and download logs.

- Verify whether `185.220.101.45` is an authorized source, such as a corporate VPN, proxy, remote office, or known administrator location.

- Verify whether `192.168.1.20` is a legitimate internal workstation or server associated with the `admin` account.

- Review `admin` account authentication history around `2026-10-03 09:00` to `09:10`, including MFA status, password changes, session creation, and lockout events.

- Determine whether the successful login at `09:04:31` used the same credentials as the failed attempts or a different authentication method.

- Investigate the `Unknown Device` event at `09:05:10`: identify the device, its owner, and whether it was authorized.

- Review file access and download logs for `Employee_Salaries.xlsx`, including destination, file size, hash, and any DLP or egress controls.

- Review database audit logs for `Employee Database` access at `09:08:22`, including queries executed and data returned.

- Check for additional activity from `185.220.101.45` against other accounts or resources.

- Check for post-incident activity after `09:10:14`, such as new accounts, privilege changes, persistence, or lateral movement.

- Interview or contact the `admin` user to determine whether the activity was authorized.

- If unauthorized, consider immediate containment: disable or reset the `admin` account, revoke active sessions, block or restrict `185.220.101.45`, and isolate affected systems.

8. Important Evidence to Verify

- Whether `185.220.101.45` is known, authorized, or associated with suspicious activity.

- Whether `192.168.1.20` is a legitimate internal host and who used it during the failed logins.

- MFA status and authentication method for the successful `admin` login.

- Whether the `admin` account password was changed, reset, or compromised.

- Details of the `Unknown Device` access: device identity, ownership, and authorization status.

- File access and download details for `Employee_Salaries.xlsx`: destination, size, hash, and DLP outcome.

- Database access details for `Employee Database`: queries, records accessed, and data volume.

- Any subsequent events after `2026-10-03 09:10:14`.

- Timezone consistency and log integrity across all sources.

- Whether other users or systems show activity from `185.220.101.45`.

9. Final Investigation Conclusion

Confirmed observations: the logs show three failed `admin` logins from `192.168.1.20`, followed by a successful `admin` login from `185.220.101.45`, then access to an `Unknown Device`, `Employee_Salaries.xlsx`, and `Employee Database`, and finally a download of `Employee_Salaries.xlsx`.

Investigative conclusion: this sequence is highly suspicious and consistent with possible unauthorized access or account compromise. The strongest evidence is the successful `admin` login from a different source IP immediately after failed attempts, followed by sensitive file and database access and a file download.

However, the supplied evidence does not confirm a malicious incident. It does not prove how the `admin` credentials were obtained, whether the activity was authorized, whether `185.220.101.45` is malicious, or whether data was exfiltrated outside the environment. This should be treated as a suspected account compromise or unauthorized access incident requiring immediate verification and containment if authorization cannot be established.

6. Investigation Limitation

This report is based on the security log evidence supplied to the investigation assistant. Root-cause findings and AI-generated conclusions are investigative hypotheses and should be verified using additional authentication, endpoint, network, and application evidence.